

-Full name: Ha Hoang PhucMSSV: 222683Class: DH22TIN07 MRI 1: Library management system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers)+ Book: main entity, managing book information.

- + DocGia: person who borrows books.

- + PhieuMuon: records borrowing/returning.

- + NhanVien: system manager (librarian, warehouse keeper...).

- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class+ Sach

- o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), thai page (string: "Available"/"On loan").

- o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

- + DocGia

- o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

- o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().+

- PhieuMuon

- o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoHenTra (date), ngoTraThucTe (date), tinhTrang (string: "On loan"/"Paid"/"Overdue").

- o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(),

- crystalPhieuPhat().+ NhanVien

- o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

- o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().).

- + NhaXuatBan

- o Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).

- o Behavior: addNXB (), capNhatNXB ().

2- Relationships: Books and Publishers: 1

book – 1 publisher

- ☐☐ Readers and Borrowing Slips: 1 reader – many borrowing slips
- ☐☐ Books and borrowing slips: 1 book – many borrowing slips
- ☐☐ Employees and Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ☐☐ DangKyTaiKhan (Customer registration).
- ☐☐ DangNhap (Sign in).
- ☐☐ See SanPham (Search / View product list).
- ☐☐ ThemVaoGioHang (Add to cart).
- ☐☐ DatHang (Order).
- ☐☐ ThanhToan (Online payment / COD).
- ☐☐ View DonHang (View order history).

Admin / NguoiQuanLy:

- ☐☐ QuanLySanPham (Add/Edit/Delete products).
- ☐☐ QuanLyDonHang (View/Confirm/Cancel order).
- ☐☐ QuanLyKhachHang (Customer information management).
- ☐☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ☐☐ ThanhToan includes XacNhanDonHang.
- ☐☐ DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- ☐☐ Customer (Customer): from register, view orders.
- ☐☐ SanPham (Product): product management.
- ☐☐ GioHang (Cart): temporarily store products before ordering.
- ☐☐ Don Hang (Order): from order, payment.
- ☐☐ ChiTietDonHang (OrderDetail): product details in the order.
- ☐☐ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- ☐☐ Admin or Personnel (inherited from User if there is a user system).

☐☐- Analyze relationships between classes (specify inheritance if any)☐ Customer and DonHang: 1 customer – many orders☐ DonHang and ChiTietDon☐☐☐Hang: 1 order – many order details (Composition)☐ ChiTietDonHang and SanPham: many simple details – 1 product☐ DonHang and ThanhTotal: 1 order - 1 payment☐ User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and Admin are both the same type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data type	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	nameMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data type	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	